JAMA Cardiology | Original Investigation

Association Between Hospital Volume and 30-Day Readmissions Following Transcatheter Aortic Valve Replacement

Sahil Khera, MD, MPH; Dhaval Kolte, MD, PhD; Tanush Gupta, MD; Andrew Goldsweig, MD; Poonam Velagapudi, MD, MS; Ankur Kalra, MD; Gilbert H. L. Tang, MD, MSc, MBA; Wilbert S. Aronow, MD; Gregg C. Fonarow, MD; Deepak L. Bhatt, MD, MPH; Herbert D. Aronow, MD, MPH; Neal S. Kleiman, MD; Michael Reardon, MD; Paul C. Gordon, MD; Barry Sharaf, MD; J. Dawn Abbott, MD

Editorial page 721 and Editor's Note page 741

Supplemental content

IMPORTANCE With the approval of transcatheter aortic valve replacement (TAVR) for patients with severe symptomatic aortic stenosis at intermediate surgical risk, TAVR volume is projected to increase exponentially in the United States. The 30-day readmission rate for TAVR was recently reported at 17.9%. The association between institutional TAVR volume and the 30-day readmission metric has not been examined.

OBJECTIVE To assess the association between hospital TAVR volume and 30-day readmission.

DESIGN, SETTING, AND PARTICIPANTS In this observational study, we used the 2014 Nationwide Readmissions Database to identify hospitals with established TAVR programs (performing at least 5 TAVRs in the first quarter of 2014). Based on annual TAVR volume, hospitals were classified as low (<50), medium (≥50 to <100), and high (≥100) volume. Rates, causes, and costs of 30-day readmissions were compared between low-, medium-, and high-volume hospitals. Data were analyzed from November to December 2016.

EXPOSURE Transcatheter aortic valve replacement.

MAIN OUTCOMES AND MEASURES Thirty-day readmissions.

RESULTS Of 129 hospitals included in this study, 20 (15.5%) were categorized as low volume, 47 (36.4%) as medium volume, and 62 (48.1%) as high volume. Of 16 252 index TAVR procedures, 663 (4.1%), 3067 (18.9%), and 12 522 (77.0%) were performed at low-, medium-, and high-volume hospitals, respectively. Thirty-day readmission rates were significantly lower in high-volume compared with medium-volume (adjusted odds ratio, 0.76; 95% CI, 0.68-0.85; $P < .001$) and low-volume (adjusted odds ratio, 0.75; 95% CI, 0.60-0.92; $P = .007$) hospitals. Noncardiac readmissions were more common in low-volume hospitals (65.6% vs 60.6% in high-volume hospitals), whereas cardiac readmissions were more common in high-volume hospitals (39.4% vs 34.4% in low-volume hospitals). There were no significant differences in length of stay and costs per readmission among the 3 groups (mean [SD], 5.5 [5.0] days vs 5.9 [7.5] days vs 6.0 [5.8] days; $P = .74$, and $13 886 [18 333] vs $14 135 [17 939] vs $13 432 [15 725]; $P = .63$, respectively).

CONCLUSIONS AND RELEVANCE We report for the first time, to our knowledge, an inverse association between hospital TAVR volume and 30-day readmissions. Lower readmission at higher-volume hospitals was associated with significantly lower cost to the health care system.

JAMA Cardiol. 2017;2(7):732-741. doi:10.1001/jamacardio.2017.1630
Published online May 11, 2017.

Author Affiliations: Author affiliations are listed at the end of this article.

Corresponding Author: J. Dawn Abbott, MD, Division of Cardiology, Department of Medicine, Warren Alpert Medical School of Brown University, 593 Eddy St, RIH APC814, Providence, RI 02903 (jabbott@lifespan.org).

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

Transcatheter aortic valve replacement (TAVR) is guideline recommended for patients with symptomatic severe aortic stenosis at prohibitive risk or "selected" patients at high conventional surgical risk.[1-6] The volume of TAVR has been increasing worldwide, with TAVR surpassing conventional surgical aortic valve replacements in some European nations.[7,8] With the recent US Food and Drug Administration approval of the Edwards Sapien XT and Sapien 3 valves (Edwards Lifesciences) for intermediate surgical risk patients, the use of TAVR is expected to increase exponentially in the United States.[9,10] Improved case selection and procedural and technical refinements coupled with operator experience have led to better TAVR outcomes in the United States; however, TAVR programs (irrespective of experience or volume) are under constant pressure to further improve outcome metrics.[11]

Readmission is a frequently used quality metric, especially in the realm of cardiovascular diseases and procedures (heart failure, acute myocardial infarction, coronary artery bypass graft surgery, and percutaneous coronary intervention).[12] The most recent data from elderly Medicare beneficiaries reported a 17.8% readmission rate for targeted conditions (acute myocardial infarction, heart failure, and pneumonia).[13] Rising health care expenditures have led hospital systems and practices to track outcomes and expenditures, especially for novel procedures. In prior studies, the 30-day readmission rate for TAVR was reported at 17.9% and high-volume TAVR hospitals have been shown to have lower in-hospital mortality, shorter length of stay (LOS), and reduced hospital costs.[14-16]

However, to our knowledge, there are no data from the US TAVR experience assessing the association between hospital TAVR volume and 30-day readmission rates. We designed this observational study to assess 30-day readmission rates at low-, medium-, and high-volume TAVR hospitals and to compare the causes and cost of readmission.

Key Points

Question Is there an association between hospital transcatheter aortic valve replacement (TAVR) volume and 30-day readmission rate?

Findings This observational study found that high-volume TAVR hospitals had significantly lower 30-day readmission rates compared with medium- and low-volume TAVR hospitals.

Meaning There exists an inverse association between hospital TAVR volume and 30-day readmission rates.

Methods

Data Source

The 2014 Nationwide Readmissions Database (NRD) was used for this study. The NRD is part of a family of databases developed by the Agency for Healthcare Research and Quality for the Healthcare Cost and Utilization Project. The 2014 NRD is constructed from the State Inpatient Databases of 22 states that are geographically dispersed and represent 51.2% of the total US resident population and 49.3% of all US hospitalizations. Unweighted, the NRD contains data from approximately 15 million discharges each year. Weighted, it estimates approximately 35 million discharges. Weighting of data by applying the DISCWT variable was done to obtain national estimates. The NRD contains clinical and nonclinical variables that support readmission analyses, while protecting the privacy of individual patients, physicians, and hospitals. This study was deemed exempt from institutional review board approval by the New York Medical College institutional review board because the NRD is publicly available and contains deidentified patient data.

Study Population

Hospitalizations for TAVR were identified using the *International Classification of Diseases, Ninth Revision, Clinical Modification* procedure codes 35.05 and 35.06. A total of 241 hospitals performed TAVRs in 2014. Hospitals with an established TAVR program (defined as performing at least 5 TAVRs in the first quarter of 2014) were identified (n = 129) and included in the present study. These hospitals performed 18 601 TAVRs in 2014. Based on the annual number of TAVRs performed, hospitals were categorized as low-volume (<50), medium-volume (≥50 to <100), and high-volume (≥100) hospitals. These cutoffs were determined based on the median TAVR volume in our study cohort (97 TAVRs per year). We categorized high-volume hospitals as those above the median and subdivided the ones below the median into 2 groups: low and medium. The Centers for Medicare and Medicaid Services do not provide cutoffs for hospital classification as low-, medium-, or high-volume TAVR hospitals. Moreover, the goal of this study was not to identify ideal TAVR volume cutoffs as TAVR is still an evolving field and institutional volumes are likely to increase in the future. We excluded hospitals that did not perform at least 5 TAVRs in the first quarter to avoid confounding data from nonestablished TAVR programs. A total of 1049 TAVRs were performed in these excluded hospitals in 2014, with a median of 3 TAVRs per center (interquartile range, 2-16). Because the primary outcome of interest was 30-day readmissions, patients discharged in December 2014 (n = 1698) were excluded owing to lack of follow-up data. We also excluded patients who died during the index hospitalization (n = 647 [in-hospital mortality = 3.8% overall and 4.3%, 4.4%, and 3.7% in low-, medium-, and high-volume hospitals, respectively]). For patients who underwent repeat TAVR within 30 days (n = 4), the second discharge record was considered a readmission. On the other hand, for patients who underwent repeat TAVR after 30 days (n = 38), the second discharge record was considered a separate index procedure. The final study sample consisted of 16 252 index TAVRs performed at 129 hospitals (**Figure 1**).

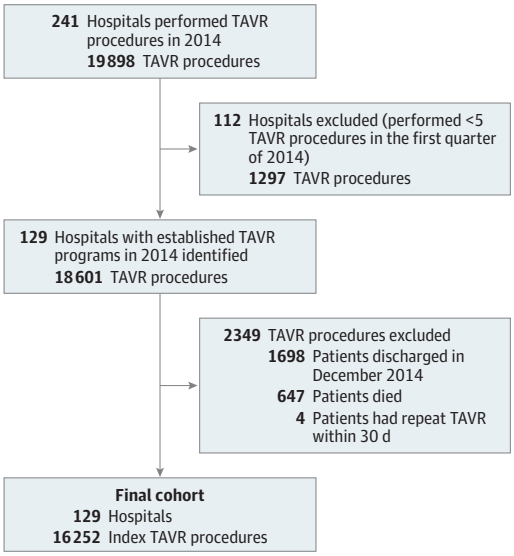
Patient and Hospital Characteristics

We included patient demographics (age, sex, expected primary payer, and median household income) and relevant comorbidities (eg, smoking, dyslipidemia, hypertension, diabetes, obesity, heart failure, known coronary artery disease, prior myocardial infarction, prior percutaneous coronary intervention, prior coronary artery bypass graft surgery, carotid artery

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

Figure 1. Study Population and Selection Criteria



TAVR indicates transcatheter aortic valve replacement.

disease, prior transient ischemic attack [TIA]/stroke, atrial fibrillation, prior permanent pacemaker, prior implantable cardioverter defibrillator, peripheral vascular disease, anemia, chronic kidney disease, chronic lung disease, liver disease, coagulopathy, dementia, depression, hypothyroidism, fluid and electrolyte disorders, other neurological disorders, pulmonary circulation disorders, and cancer). Hospital characteristics, such as location, teaching status, and bed size, were also examined. Other variables extracted were TAVR access site (endovascular [EV] vs transapical [TA]), in-hospital procedures (coronary angiography, percutaneous coronary intervention, and mechanical circulatory support), and in-hospital complications (conversion to surgical aortic valve replacement, complete heart block, permanent pacemaker placement, TIA/stroke, acute myocardial infarction, cardiogenic shock, cardiac arrest, acute kidney injury, major bleeding, and vascular complications). The Healthcare Cost and Utilization Project Clinical Classification Software and *International Classification of Diseases, Ninth Revision, Clinical Modification* codes used to define these variables are listed in eTable 1 in the Supplement.

Outcomes Measured

The primary outcome studied was 30-day all-cause readmission. Readmissions were identified according to the methods outlined by the Healthcare Cost and Utilization Project.[17] For patients who had multiple readmissions within 30 days, only the first readmission was included. Transfer to another hospital was not considered a readmission. To determine the cause of readmission, the primary diagnosis of each readmission record was independently reviewed by 2 authors (S.K. and D.K.) and grouped into clinically relevant categories, as previously described.[16] Discrepancies were resolved by mutual agreement. eTable 2 in the Supplement lists the primary diagnosis categories and the corresponding Clinical Classification Software and/or *International Classification of Diseases, Ninth Revision, Clinical Modification* codes. Secondary outcomes examined were LOS and hospital costs for the readmission episode. Costs were inflation adjusted using the US Bureau of Labor Statistics Consumer Price Index, with 2016 as the index base.[18]

Statistical Analysis

Patient demographics, comorbidities, hospital characteristics, in-hospital procedures, and in-hospital complications were compared between low-, medium-, and high-volume hospitals using the Pearson χ^2 test for categorical variables and 1-way analysis of variance for continuous variables. To examine differences in 30-day readmissions among low-, medium-, and high-volume hospitals, multivariable logistic regression models were constructed using generalized estimating equations to account for clustering of outcomes within hospitals. The models included all variables listed in **Table 1** (except hospital location because 99.6% were in an urban location) as covariates. We also included the discharge quarter variable in our regression model to partially account for the learning curve associated with TAVR. We performed exploratory analyses comparing 30-day readmission rates between low/medium- vs high-volume hospitals for each quarter of 2014. We also performed a sensitivity analysis using tertiles of annual hospital TAVR volume as cutoff (tertile 1: <74, tertile 2: ≥74 to <143, and tertile 3: ≥143) to determine whether the association between hospital TAVR volume and 30-day readmission rates remained significant. We also examined the association between TAVR volume and 90-day readmission rates.

Data were complete for all variables except primary expected payer (0.1% missing) and median household income (1.3% missing). Missing values were replaced with the dominant category for these variables. Data on cost were missing for 308 index hospitalizations and 26 readmissions. Therefore, results of cost analysis are based on a sample size of 15 944 index hospitalizations and 2641 readmissions.

Statistical analysis was performed with IBM SPSS Statistics version 20.0 (IBM Corp). All *P* values were 2-sided, with a significance threshold of $P < .05$. Categorical variables are expressed as percentages and continuous variables as mean (SD) or median (interquartile range) as appropriate. Odds ratios (ORs) and 95% CIs are used to report the results of regression analysis.

Results

Of 129 TAVR hospitals included in this study, 20 (15.5%) were categorized as low volume, 47 (36.4%) as medium volume, and 62 (48.1%) as high volume. Of 16 252 index TAVR procedures, 663 (4.1%), 3067 (18.9%), and 12 522 (77.0%) were performed at low-, medium-, and high-volume hospitals, respectively.

Baseline Characteristics

Table 1 summarizes the baseline characteristics, in-hospital procedures, and in-hospital outcomes of patients who underwent TAVR at low-, medium-, and high-volume hospitals and who were discharged alive. Compared with patients undergo-

© 2017 American Medical Association. All rights reserved.
Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

Table 1. Baseline Characteristics and In-Hospital Outcomes of Patients Undergoing TAVR at Low-, Medium-, and High-Volume Hospitals

Characteristic	No. (%)				P Value
		Hospital TAVR Volume			
	Overall	Low (n = 663)	Medium (n = 3067)	High (n = 12 522)	
Patient Characteristics					
Age, mean (SD), y	81.2 (8.3)	82.2 (7.6)	81.1 (8.5)	81.2 (8.2)	.01
<85	9226 (56.8)	352 (53.2)	1739 (56.7)	7135 (57.0)	.16
≥85	7026 (43.2)	310 (46.8)	1328 (43.0)	5388 (43.0)	
Women	7510 (46.2)	337 (50.8)	1381 (45.0)	5792 (46.3)	.02
Primary expected payer					
Medicare	14 892 (91.6)	590 (89.0)	2839 (92.5)	11 463 (91.5)	.001
Medicaid	148 (0.9)	13 (2.0)	34 (1.1)	101 (0.8)	
Private	915 (5.6)	46 (6.9)	158 (5.1)	711 (5.7)	
Uninsured	62 (0.4)	<10 (0.6)	11 (0.4)	47 (0.4)	
Other	236 (1.5)	10 (1.5)	26 (0.8)	200 (1.6)	
Median household income, percentile					
0-25th	2843 (17.5)	151 (22.8)	627 (20.4)	2065 (16.5)	<.001
26th-50th	4271 (26.3)	166 (25.0)	820 (26.7)	3285 (26.2)	
51st-75th	4149 (25.5)	157 (23.7)	790 (25.8)	3202 (25.6)	
76th-100th	4989 (30.7)	189 (28.5)	830 (27.1)	3970 (31.7)	
Comorbidities					
Smoking	4891 (30.1)	189 (28.5)	869 (28.3)	3833 (30.6)	.03
Dyslipidemia	10 388 (63.9)	430 (64.9)	1924 (62.7)	8034 (64.2)	.30
Hypertension	12 967 (79.8)	529 (79.9)	2384 (77.7)	10 054 (80.3)	.006
Diabetes	5680 (35.0)	255 (38.5)	1121 (36.6)	4304 (34.4)	.01
Obesity	2611 (16.1)	104 (15.7)	464 (15.1)	2043 (16.3)	.26
Heart failure	12 218 (75.2)	469 (70.7)	2288 (74.6)	9461 (75.6)	.01
Known CAD	11 130 (68.5)	450 (67.9)	1924 (62.7)	8757 (69.9)	<.001
Prior MI	1987 (12.2)	76 (11.5)	359 (11.7)	1552 (12.4)	.48
Prior PCI	3213 (19.8)	150 (22.6)	622 (20.3)	2441 (19.5)	.11
Prior CABG	3454 (21.3)	147 (22.2)	685 (22.3)	2622 (20.9)	.20
Carotid artery disease	1108 (6.8)	31 (4.7)	205 (6.7)	872 (7.0)	.07
Prior TIA/stroke	2091 (12.9)	104 (15.7)	438 (14.3)	1549 (12.4)	.002
Atrial fibrillation	7218 (44.4)	305 (46.1)	1400 (45.6)	5513 (44.0)	.18
Prior PPM	1803 (11.1)	65 (9.8)	362 (11.8)	1376 (11.0)	.24
Prior *ICD*	461 (2.8)	22 (3.3)	82 (2.7)	357 (2.9)	.65
Peripheral vascular disease	3619 (28.5)	192 (29.0)	818 (26.7)	3619 (28.9)	.048
Anemia	4036 (24.8)	210 (31.7)	744 (24.3)	3082 (24.6)	<.001
Chronic kidney disease	5858 (36.0)	242 (36.5)	1046 (34.1)	4570 (36.5)	.046
Chronic lung disease	5459 (33.6)	227 (34.2)	897 (29.2)	4335 (34.6)	<.001
Liver disease	514 (3.2)	22 (3.3)	102 (3.3)	390 (3.1)	.81
Coagulopathy	3753 (23.1)	171 (25.8)	656 (21.4)	2926 (23.4)	.02
Dementia	1328 (8.2)	71 (10.7)	239 (7.8)	1018 (8.1)	.04
Depression	1237 (7.6)	51 (7.7)	223 (7.3)	963 (7.7)	.73
Hypothyroidism	3113 (19.2)	157 (23.7)	542 (17.7)	2414 (19.3)	.001
Fluid and electrolyte disorders	4602 (28.3)	230 (34.7)	695 (22.7)	3677 (29.4)	<.001
Other neurological disorders	951 (5.9)	56 (8.4)	167 (5.4)	728 (5.8)	.01
Pulmonary circulation disorders	922 (5.7)	35 (5.3)	188 (6.1)	699 (5.6)	.46
Cancer	660 (4.1)	21 (3.2)	109 (3.6)	530 (4.2)	.12
No. of Elixhauser comorbidities, mean (SD)	4.4 (1.9)	4.7 (1.9)	4.2 (1.9)	4.4 (1.9)	<.001
≤4	8822 (54.3)	319 (48.1)	1839 (59.9)	6664 (53.2)	<.001
>4	7431 (45.7)	344 (51.9)	1229 (40.1)	5858 (46.8)	

(continued)

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

Table 1. Baseline Characteristics and In-Hospital Outcomes of Patients Undergoing TAVR at Low-, Medium-, and High-Volume Hospitals (continued)

Characteristic	No. (%)				
		Hospital TAVR Volume			
	Overall	Low (n = 663)	Medium (n = 3067)	High (n = 12 522)	*P* Value
Hospital Characteristics					
Bed size					
Small	750 (4.6)	29 (4.4)	248 (8.1)	473 (3.8)	<.001
Medium	1816 (11.2)	198 (29.9)	477 (15.5)	1141 (9.1)	
Large	13 686 (84.2)	435 (65.7)	2343 (76.4)	10 908 (87.1)	
Location					
Rural	59 (0.4)	0 (0)	59 (1.9)	0 (0)	<.001
Urban	16 193 (99.6)	663 (100.0)	3008 (98.1)	12 522 (100.0)	
Teaching status					
Nonteaching	1048 (6.4)	71 (10.7)	501 (16.3)	476 (3.8)	<.001
Teaching	15 205 (93.6)	592 (89.3)	2567 (83.7)	12 046 (96.2)	
In-hospital Procedures					
TAVR access					
Endovascular	13 546 (83.3)	491 (74.1)	2357 (76.9)	10 698 (85.4)	<.001
Transapical	2706 (16.7)	172 (25.9)	710 (23.1)	1824 (14.6)	
Coronary angiography	3252 (20.0)	103 (15.5)	477 (15.6)	2672 (21.3)	<.001
PCI	532 (3.3)	26 (3.9)	70 (2.3)	436 (3.5)	.002
Mechanical circulatory support	213 (1.3)	13 (2.0)	40 (1.3)	160 (1.3)	.32
In-hospital outcomes					
Surgical AVR	43 (0.3)	<10 (0.2)	<10 (0.3)	34 (0.3)	.84
Complete heart block	1703 (10.5)	63 (9.5)	258 (8.4)	1382 (11.0)	<.001
PPM placement	1892 (11.6)	68 (10.3)	296 (9.7)	1528 (12.2)	<.001
TIA/stroke	370 (2.3)	17 (2.6)	87 (2.8)	266 (2.1)	.053
AMI	395 (2.4)	24 (3.6)	61 (2.0)	310 (2.5)	.04
Cardiogenic shock	318 (2.0)	17 (2.6)	58 (1.9)	243 (1.9)	.51
Cardiac arrest	400 (2.5)	27 (4.1)	83 (2.7)	290 (2.3)	.01
AKI	2690 (16.6)	111 (16.8)	472 (15.4)	2107 (16.8)	.16
Major bleeding	2146 (13.2)	101 (15.2)	369 (12.0)	1676 (13.4)	.04
Vascular complications	599 (3.7)	27 (4.1)	124 (4.0)	448 (3.6)	.41
LOS, mean (SD), d	7.9 (7.9)	9.4 (9.8)	7.9 (7.7)	7.8 (7.8)	<.001
≤5	8092 (49.8)	288 (43.4)	1525 (49.7)	6279 (50.1)	.003
>5	8161 (50.2)	375 (56.6)	1543 (50.3)	6243 (49.9)	
Discharge disposition					
Home (self-care)	5710 (35.1)	197 (29.7)	1195 (39.0)	4318 (34.5)	<.001
Short-term hospital	173 (1.1)	<10 (1.1)	57 (1.9)	109 (0.9)	
Skilled nursing facility	4239 (26.1)	215 (32.4)	788 (25.7)	3236 (25.8)	
Home health care	6119 (37.6)	245 (36.9)	1024 (33.4)	4850 (38.7)	
Total hospital cost, $	57 812 (30 596)	63 942 (34 704)	61 884 (31 424)	56 492 (30 026)	<.001

Abbreviations: AKI, acute kidney injury; AMI, acute myocardial infarction; AVR, aortic valve replacement; CABG, coronary artery bypass graft; CAD, coronary artery disease; *ICD*, *International Classification of Diseases*; LOS, length of stay; MI, myocardial infarction; PCI, percutaneous coronary intervention; PPM, prior permanent pacemaker; SD, standard deviation; TAVR, transcatheter aortic valve replacement; TIA, transient ischemic attack.

ing TAVR at low-volume hospitals, those at high-volume hospitals were younger, less likely to be women, and had higher median household income. Patients undergoing TAVR at medium- and high-volume hospitals had a lower burden of comorbidities, especially lower prevalence of diabetes, prior TIA/stroke, anemia, fluid and electrolyte disorders, and neurological disorders compared with those at low-volume hospitals. Patients at medium- and high-volume hospitals were more likely to undergo TAVR via EV access as opposed to more TA TAVRs at low-volume hospitals.

Rates of complete heart block and permanent pacemaker placement were higher at medium- and high-volume hospitals, whereas low-volume hospitals had a higher incidence of other in-hospital complications such as acute myocardial infarction, cardiac arrest, and major bleeding. Medium- and high-volume hospitals had shorter LOS, lower costs of index hospitalization, and lower rates of discharge to skilled nursing facilities compared with low-volume hospitals.

Incidence of 30-Day Readmissions

The overall incidence of 30-day readmissions was 16.4%. The median time to readmission was 9 days (interquartile range, 5-17 days). There was no significant difference in 30-day readmission rates between low- and medium-volume hospitals

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

(19.5% vs 19.0%; unadjusted OR, 0.97; 95% CI, 0.79-1.21 and adjusted OR, 0.98; 95% CI, 0.78-1.23, respectively) (**Figure 2**). However, high-volume hospitals had significantly lower 30-day readmission rates (15.6%) compared with medium-volume (adjusted OR, 0.76; 95% CI, 0.68-0.85; $P < .001$) and low-volume (adjusted OR, 0.75; 95% CI, 0.60-0.92; $P = .007$) hospitals (**Table 2**).

For sensitivity analysis, hospitals were divided into tertiles of annual TAVR volume. Thirty-day readmission rates were similar to our absolute volume cutoffs (19.1%, 18.3%, and 15.2% for lowest, middle, and highest tertiles, respectively). Hospitals in the highest-volume tertile were 28% less likely to have a 30-day readmission compared with those in the lowest tertile (adjusted OR, 0.72; 95% CI, 0.63-0.82) (eTable 3 and eFigure 1 in the Supplement). The TAVR volume-readmissions association remained consistent for 90-day readmission rates (eFigure 2 in the Supplement). Results of exploratory analyses comparing 30-day readmission rates between low/medium- vs high-volume hospitals for each quarter of 2014 are provided in eTable 4 in the Supplement.

Causes of 30-Day Readmissions

Of 2667 readmissions, 1619 (61.7%) were due to noncardiac causes and 1048 (39.3%) were due to cardiac causes. Compared with medium- and high-volume hospitals, low-volume hospitals had a numerically lower proportion of cardiac readmissions (34.4% vs 39.9% vs 39.4%, respectively) and higher proportion of noncardiac readmissions (65.6% vs 60.1% vs 60.6%, respectively); however, this difference was not statistically significant ($P = .36$) (**Figure 3**A). Among the noncardiac causes of readmission, infection, respiratory, endocrine/metabolic, renal, and trauma problems were more common in low-volume hospitals, whereas gastrointestinal and TIA/stroke issues were more common in medium- and high-volume hospitals. Among the cardiac causes, heart failure, arrhythmias, and conduction disorders were more common in medium- and high-volume hospitals, whereas coronary artery disease, chest pain, and syncope were more common in low-volume hospitals (Figure 3B). We further categorized causes of readmissions as TAVR-related (cardiac plus noncardiac [bleeding, TIA/stroke, acute kidney injury, postoperative infections, and infections due to cardiac/vascular device]) and non-TAVR-related (all other noncardiac causes). Low-volume hospitals had a higher proportion of non-TAVR-related readmissions compared with medium- and high-volume hospitals (54.3% vs 45.7% vs 46.9%), but this difference was not statistically significant ($P = .21$) (eFigure 3 in the Supplement).

LOS and Costs of 30-Day Readmissions

There were no significant differences in the mean LOS and costs of readmission between low-, medium-, and high-volume hospitals (mean [SD], 5.5 [5.0] days vs 5.9 [7.5] days vs 6.0 [5.8] days; $P = .74$, and $13 886 [18 333] vs $14 135 [17 939] vs $13 432 [15 725]; $P = .63$, respectively). More importantly, we calculated that if the high-volume hospitals had similar readmission rates as low-volume hospitals (ie, 19.5% [2442] instead of 15.6% [1957] of readmissions), the excess readmissions would have increased the annual health care expenditure by more than $6.5 million.

Discussion

Transcatheter aortic valve replacement is the accepted standard of care for appropriately selected patients with symptomatic severe aortic stenosis who are at prohibitive or high surgical risk. The Centers for Medicare and Medicaid Services requires a hospital to perform at least 50 surgical aortic valve replacements (including for at least 10 high-risk patients), employ 2 or more physicians with cardiac surgery privileges, and perform 1000 or more coronary angiograms (≥400 percutaneous coronary interventions) prior to approval as a TAVR site.[19] Preexisting hospital volume (interventional and surgical) and available physician resources play a key role not only in the initial approval, but also on program maintenance and reimbursements. It underlines the importance given to hospital volumes in predicting patient outcomes.

In our analysis, we report several novel findings: (1) there was an inverse association between hospital TAVR volume and 30-day readmission rates; (2) noncardiac readmissions were

Figure 2. 30-Day Readmission Rates in Low-, Medium-, and High-Volume Transcatheter Aortic Valve Replacement Hospitals

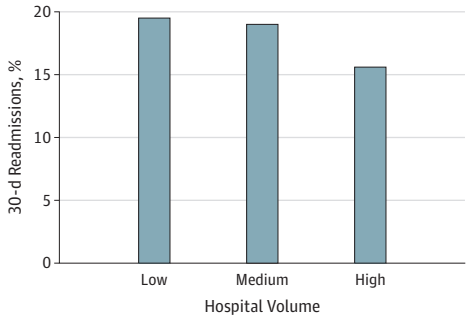


Table 2. 30-Day Readmission Rates According to Annual Hospital TAVR Volume

Measure	Annual Hospital TAVR Volume		
	Low (<50) (n = 663)	Medium (≥50 to <100) (n = 3067)	High (≥100) (n = 12 522)
No. (%)	129 (19.5)	582 (19.0)	1957 (15.6)
OR (95% CI)			
Unadjusted[a]	1 [Reference]	0.97 (0.79-1.21)	0.76 (0.62-0.92)
Adjusted[b]	1 [Reference]	0.98 (0.78-1.23)	0.75 (0.60-0.92)

Abbreviations: OR, odds ratio; TAVR, transcatheter aortic valve replacement.

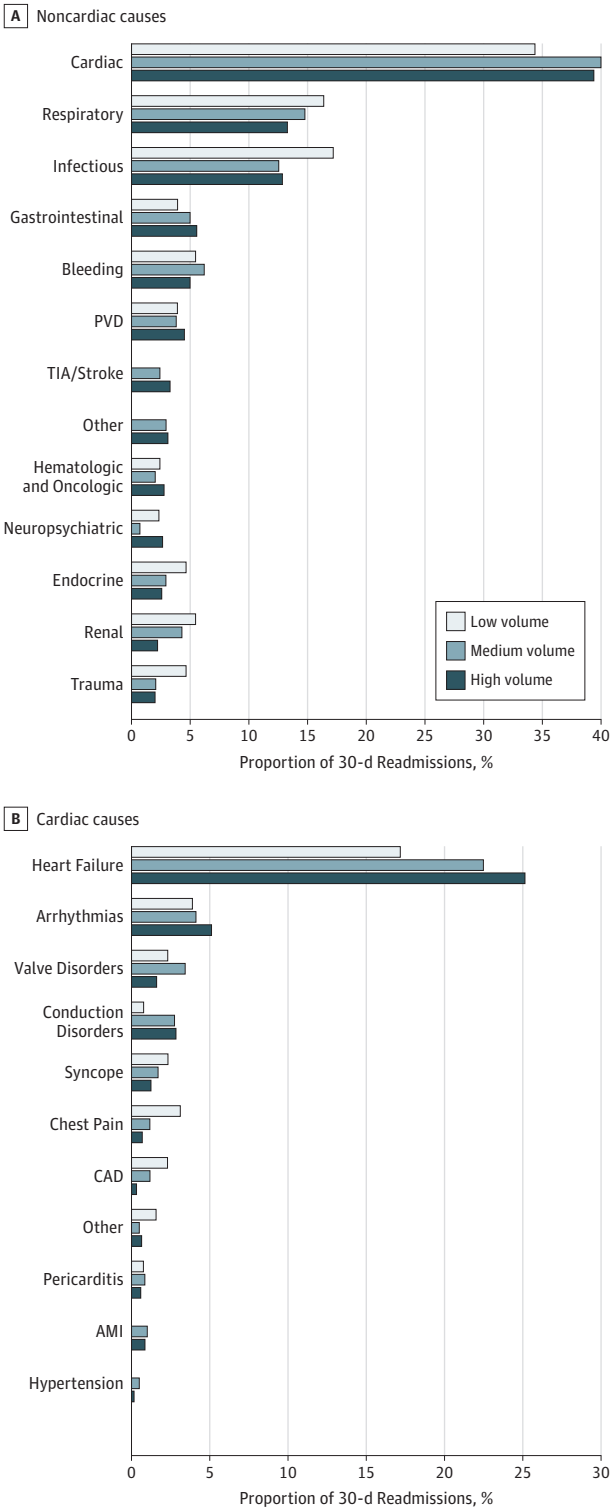
[a] High vs medium: OR, 0.78 (95% CI, 0.70-0.86).

[b] High vs medium: OR, 0.76 (95% CI, 0.68-0.85).

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

Figure 3. Causes of 30-Day Readmissions After Transcatheter Aortic Valve Replacement



The graphs show noncardiac (A) and cardiac (B) causes of 30-day readmissions in low-, medium-, and high-volume hospitals. AMI indicates acute myocardial infarction; CAD, coronary artery disease; PVD, peripheral vascular disease; and TIA, transient ischemic attack.

more common in low-volume hospitals, whereas cardiac readmissions were more common in high-volume hospitals; and (3) there were no differences between mean LOS and costs of readmissions between high-, medium-, and low-volume hospitals. Although the cost of all readmissions was similar, lower readmission rates at high-volume hospitals equate to lower health care costs and substantial savings.

Readmission is an important metric not only for reimbursement purposes, but also for tracking the outcomes of any novel procedure. The 30-day readmission rate for 6 surgical procedures (coronary artery bypass graft surgery, pulmonary lobectomy, EV repair of abdominal aortic aneurysm, open repair of abdominal aortic aneurysm, colectomy, and hip replacement) using Medicare data was 13.1%.[20] More importantly, for these surgical procedures, the highest-volume hospitals had the lowest readmissions rates when compared with the lowest-volume hospitals (12.7% vs 16.8%; $P < .001$).[20] The previously reported TAVR 30-day readmission rates have been high and variable: 17.4% (Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy [TVT] Registry), 17.9% (NRD), 18.8% (New York's Cardiac Surgery Reporting System), and 20.9% (Medicare data).[16,21-23]

We report a 25% lower readmission rate in high-volume hospitals (≥100 TAVRs annually) compared with low-volume hospitals (<50 TAVRs annually). Prior to this study, to our knowledge, there were no data on the association between TAVR hospital volume and 30-day readmission rates. Murugiah et al[22] used Medicare data to understand the relationships between hospitals' all-cause readmission rate and their 30-day TAVR patient readmission rate. In their analysis, the odds of 30-day readmission was 1.41 (95% CI, 1.37-1.44) when the patient was treated at a hospital 1 SD above the national average for 30-day all-cause readmissions vs a hospital 1 SD below the national average for 30-day all-cause readmissions.[22] The authors postulated that hospital and system factors may play a key role in this observed variation. However, one of the limitations of their study was the lack of assessment of the relationship between hospital TAVR volume and outcomes.

The inverse association between hospital TAVR volume and 30-day readmission rates seen in our study is potentially of concern. Low-volume hospitals were more likely to operate on patients with a higher number of comorbidities compared with high-volume hospitals and were more likely to use the TA approach. Transapical TAVR is associated with poorer short- and intermediate-term mortality, increased use of skilled nursing care facilities, longer hospital stays, and readmissions when compared with transfemoral TAVR.[16,24-26] Transapical TAVR is associated with lower risk for vascular complications and is useful in patients with severe extensive peripheral vascular disease. However, in our analysis, preexisting peripheral vascular disease was similar in both groups. It is possible that the TA approach was used more frequently at low-volume sites owing to physician preference rather than patient factors alone. Alternatively, because these data were collected at the time of rapid technology evolution and development of lower profile systems, there is a possibility that higher-volume hospitals had greater access to newer technological advancements in EV access. Higher rates of complete heart block and permanent pacemaker placement at medium- and

© 2017 American Medical Association. All rights reserved.
Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

high-volume hospitals may be due to valve selection or a lower threshold to place a pacemaker during the index hospitalization rather than specific approach (TA TAVR vs EV TAVR).[5,27,28] Patient population and community resources can also affect the readmission rate and are difficult to assess from this study.[29]

The British Columbia TAVR program recently reported its success by using regional systems of care and demonstrated shorter median hospital stays (3 days vs 5 days in TVT high-risk and TVT inoperable registries) and higher rates of discharge to home (94% compared with 67% in TVT high-risk and 70% in TVT inoperable registries).[30] Regionalization of TAVR care may improve outcomes including lowering readmission rates. For example, referrals for extremely complex patients and TA TAVRs can be made to a single high-volume site in the region or county, whereas the less complex TAVR cases using the EV approach may be performed at lower-volume hospitals. Although our data are hypothesis generating, they do not provide sufficient argument to advocate for regionalization of TAVR care in the United States as TAVR is a maturing field. The procedure was evolving at the time of data acquisition. There is a learning curve for all new procedures and lower-volume hospitals in 2014 that may have been on the learning curve could now represent the high-volume hospitals.[31,32] It took 26 cases to achieve a consistently low risk for 30-day major adverse events according to a post-hoc analysis of the Placement of Aortic Transcatheter Valves 1 Trial.[31] Learning curve is more pertinent to low-volume hospitals and we tried to partially account for it by adjusting for discharge quarter in our multivariate models. We observed a learning curve effect, ie, difference in readmission rates between low/medium- and high-volume hospitals narrowed with increasing experience (first quarter to fourth quarter); however, the volume-readmission association still remained significant. The decline in readmission rates per quarter was greater for low/medium-volume hospitals (supporting a learning curve effect); however, even the high-volume hospitals showed a significant decline in readmission rates per quarter (supporting the inverse association between volume and readmissions even in highly experienced hospitals). As both investigational and commercial TAVR implants were included in the NRD databases, it is possible that the higher-volume hospitals had greater access to EV technologies leading to greater use of EV TAVR.

In our previously published study,[16] the most common causes of readmission after TAVR were noncardiac (61.8%). The predictors of 30-day readmissions were longer LOS, acute kidney injury, chronic kidney disease, TA TAVR, more than 4 Elixhauser comorbidities, chronic lung disease, and discharge to a skilled nursing facility.[16] Higher respiratory and infectious etiology of noncardiac readmissions at low-volume hospitals seen in the present study may be due to prolonged intensive care unit stays (owing to more TA TAVRs), greater use of general anesthesia, and surgical femoral artery cut down for transfemoral TAVR cases.[33,34] Thirty-day readmissions due to heart failure and arrhythmias were more common at high-volume hospitals.

Potentially, the drivers of the observed inverse hospital volume–30-day readmissions association are patient comorbidities, TAVR-approach selection, proficiency in EV TAVRs by more experienced operators at higher-volume hospitals (operator factors), and better postoperative care coordination at higher-volume hospitals. It is difficult to control the composition of patient population and local community resources for the hospital systems, but hospital volume, health care systems, physician expertise, and postoperative care coordination are modifiable targets for standardizing outcomes. As more TAVR programs open in the country catering to intermediate- to higher-risk subgroups, these data become even more meaningful in guiding appropriate strategies to reduce readmission rates and health care costs.

Limitations

The NRD is an administrative database and information on valve type and size, echocardiographic variables, postprocedural paravalvular leak, individual patient risk scores, and medication use are not available. Society of Thoracic Surgeons scores were not available in NRD databases. Endovascular TAVR access includes all percutaneous approaches and we were unable to differentiate between femoral, direct aortic, subclavian, iliac, caval, and carotid access using the NRD databases. Readmissions across states are not captured by NRD. Post-TAVR patients will rarely be admitted to an out-of-state hospital, making this limitation relatively inconsequential. However, 30-day readmissions have been reported using administrative databases in prior studies.[16,22,23] Mortality after discharge is not captured in the NRD and this is a limitation as hospitals with high mortality rates may have a lower readmission rate. However, in-hospital mortality was lowest in high-volume hospitals and we expect this to hold true even after discharge. Residual measured and unmeasured confounding may be present in the adjusted analyses. Causes of readmission may be subject to misattribution; however, the NRD is robust in evaluating noncardiac causes of readmissions and cost analysis when compared with other available clinical registries.

Conclusions

Using a large national readmission database, we report an inverse association between hospital TAVR volume and 30-day readmissions. The reason for this inverse association is likely multifactorial and depends on patient factors, operator factors, hospital systems, and community factors. Low-volume hospitals had a higher proportion of noncardiac readmissions (infections and respiratory) compared with high-volume hospitals, likely driven by patient factors and procedural approach selection. There were no differences in LOS and cost of readmissions between low-, medium-, and high-volume hospitals. Lower readmission rates at high-volume hospitals substantially reduce health care expenditure. As new TAVR programs open across the country, these data will guide policy makers to identify targets for optimizing and standardizing TAVR outcomes across hospitals.

ARTICLE INFORMATION

Accepted for Publication: April 13, 2017.

Published Online: May 11, 2017. doi:10.1001/jamacardio.2017.1630

Author Affiliations: New York Medical College, Valhalla (Khera, W. S. Aronow); Brown University, Providence, Rhode Island (Kolte, Goldsweig,

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

Velagapudi, H. D. Aronow, Gordon, Sharaf, Abbott); Montefiore Medical Center, Albert Einstein College of Medicine, Bronx, New York (Gupta); Houston Methodist DeBakey Heart and Vascular Center, Houston Methodist Hospital, Houston, Texas (Kalra, Kleiman, Reardon); Weill Cornell Medical College, New York, New York (Kalra, Kleiman, Reardon); Mount Sinai Health System, New York, New York (Tang); University of California–Los Angeles (Fonarow); Associate Editor for Health Care Quality and Guidelines, *JAMA Cardiology* (Fonarow); Brigham and Women's Hospital Heart & Vascular Center, Harvard Medical School, Boston, Massachusetts (Bhatt).

Author Contributions: Drs Khera and Kolte had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. Drs Khera and Kolte contributed equally to the study.
Study concept and design: Khera, Kolte, Abbott.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: Khera, Kolte.
Critical revision of the manuscript for important intellectual content: All authors.
Statistical analysis: Khera, Kolte.
Administrative, technical, or material support: Khera, Kolte, Gupta, Abbott.
Study supervision: Khera, Kolte, Tang, W. S. Aronow, Fonarow, Kleiman, Bhatt, H. D. Aronow, Reardon, Gordon, Sharaf, Abbott.

Conflict of Interest Disclosures: All authors have completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest. Dr Tang has received personal fees from Edwards Lifesciences and Medtronic. Dr Bhatt has received grants from Amarin, AstraZeneca, Bristol-Myers Squibb, Eisai, Ethicon, Medtronic, Sanofi Aventis, The Medicines Company, Roche, Pfizer, Forest Laboratories, Ischemix, Amgen, Eli Lilly, Chiesi, and Ironwood; personal fees from Duke Clinical Research Institute (data monitoring committee [DMC]), Mayo Clinic (DMC), Population Health Research Institute (DMC, clinical trial steering committee for a trial funded by Bayer), Belvoir Publications (editor in chief, *Harvard Heart Letter*), Slack Publications (chief medical editor, *Cardiology Today's Intervention*), WebMD (CME steering committees), Elsevier (advisory board, Elsevier Practice Update Cardiology), HMP Communications (editor in chief, *Journal of Invasive Cardiology*), Harvard Clinical Research Institute (clinical trial steering committee for trial funded by Boehringer Ingelheim; DMC chair for a trial funded by St Jude), Cleveland Clinic (DMC), and *Journal of the American College of Cardiology* (guest editor; associate editor); other funding from FlowCo, PLx Pharma, Takeda, Medscape Cardiology (advisory board), Regado Biosciences (advisory board), Boston VA Research Institute (board of directors), *Clinical Cardiology* (deputy editor), Veterans Administration (chair, VA Cardiovascular Assessment, Reporting, and Tracking System Program; research and publications committee), St Jude Medical (site coinvestigator), Biotronik (site coinvestigator), Cardax (advisory board), American College of Cardiology (chair, NCDR-ACTION Registry steering committee), Boston Scientific (site coinvestigator), and Elsevier (editor, *Cardiovascular Intervention: A Companion to Braunwald's Heart Disease*); personal fees and nonfinancial support from the American College of Cardiology (senior associate editor, Clinical Trials and News, ACC.org), and Society of Cardiovascular Patient Care (board of directors; secretary/treasurer); and nonfinancial support from the American Heart Association. Dr Reardon has served as a consultant for Medtronic. No other disclosures were reported.

Disclaimer: Dr Fonarow is the Associate Editor for Health Care Quality and Guidelines of *JAMA Cardiology*, but he was not involved in any of the decisions regarding review of the manuscript or its acceptance.

Meeting Presentation: This study was presented as a "Best of the Best" Abstract at the Society for Cardiovascular Angiography and Interventions Scientific Sessions; May 11, 2017; New Orleans, Louisiana.

REFERENCES

1. Leon MB, Smith CR, Mack M, et al; PARTNER Trial Investigators. Transcatheter aortic-valve implantation for aortic stenosis in patients who cannot undergo surgery. *N Engl J Med*. 2010;363(17):1597-1607.

2. Smith CR, Leon MB, Mack MJ, et al; PARTNER Trial Investigators. Transcatheter versus surgical aortic-valve replacement in high-risk patients. *N Engl J Med*. 2011;364(23):2187-2198.

3. Adams DH, Popma JJ, Reardon MJ, et al; US CoreValve Clinical Investigators. Transcatheter aortic-valve replacement with a self-expanding prosthesis. *N Engl J Med*. 2014;370(19):1790-1798.

4. Tang GH, Lansman SL, Panza JA. Beyond PARTNER: appraising the evolving trends and outcomes in transcatheter aortic valve replacement. *Cardiol Rev*. 2015;23(1):1-10.

5. Tang GH, Lansman SL, Cohen M, et al. Transcatheter aortic valve replacement: current developments, ongoing issues, future outlook. *Cardiol Rev*. 2013;21(2):55-76.

6. Nishimura RA, Otto CM, Bonow RO, et al; ACC/AHA Task Force Members. 2014 AHA/ACC guideline for the management of patients with valvular heart disease: executive summary: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines. *Circulation*. 2014;129(23):2440-2492.

7. Mylotte D, Osnabrugge RL, Windecker S, et al. Transcatheter aortic valve replacement in Europe: adoption trends and factors influencing device utilization. *J Am Coll Cardiol*. 2013;62(3):210-219.

8. Eggebrecht H, Mehta RH. Transcatheter aortic valve implantation (TAVI) in Germany 2008-2014: on its way to standard therapy for aortic valve stenosis in the elderly? *EuroIntervention*. 2016;11(9):1029-1033.

9. US Food and Drug Administration. FDA news release: FDA approves expanded indication for two transcatheter heart valves for patients at intermediate risk for death or complications associated with open-heart surgery. http://www.fda.gov/NewsEvents/Newsroom/PressAnnouncements/ucm517281.htm. Accessed December 18, 2016.

10. Leon MB, Smith CR, Mack MJ, et al; PARTNER 2 Investigators. Transcatheter or surgical aortic-valve replacement in intermediate-risk patients. *N Engl J Med*. 2016;374(17):1609-1620.

11. Beohar N, Kirtane AJ, Blackstone E, et al. Trends in complications and outcomes of patients undergoing transfemoral transcatheter aortic valve replacement: experience from the PARTNER Continued Access Registry. *JACC Cardiovasc Interv*. 2016;9(4):355-363.

12. Medicare.gov. 30-Day unplanned readmission and death measures. https://www.medicare.gov/hospitalcompare/Data/30-day-measures.html. Accessed December 18, 2016.

13. Zuckerman RB, Sheingold SH, Orav EJ, Ruhter J, Epstein AM. Readmissions, observation, and the Hospital Readmissions Reduction Program. *N Engl J Med*. 2016;374(16):1543-1551.

14. Badheka AO, Patel NJ, Panaich SS, et al. Effect of hospital volume on outcomes of transcatheter aortic valve implantation. *Am J Cardiol*. 2015;116(4):587-594.

15. Mack MJ, Brennan JM, Brindis R, et al; STS/ACC TVT Registry. Outcomes following transcatheter aortic valve replacement in the United States. *JAMA*. 2013;310(19):2069-2077.

16. Kolte D, Khera S, Sardar MR, et al. Thirty-day readmissions after transcatheter aortic valve replacement in the United States: insights from the Nationwide Readmissions Database. *Circ Cardiovasc Interv*. 2017;10(1):e004472.

17. US Agency for Healthcare Research and Quality. HCUP methods series: overview of key readmission measures and methods: report 2012-04. https://www.hcup-us.ahrq.gov/reports/methods/2012_04.pdf. Published 2012. Accessed December 18, 2016.

18. US Bureau of Labor Statistics, Division of Consumer Prices and Price Indexes. Top picks: price indexes. https://data.bls.gov/cgi-bin/surveymost?bls. Accessed December 18, 2016.

19. US Department of Health and Human Services, Centers for Medicare and Medicaid Services. Transcatheter Aortic Valve Replacement (TAVR) hospital program volume requirements. https://www.cms.gov/Outreach-and-Education/Medicare-Learning-Network-MLN/MLNMattersArticles/downloads/SE1515.pdf. Accessed December 18, 2016.

20. Tsai TC, Joynt KE, Orav EJ, Gawande AA, Jha AK. Variation in surgical-readmission rates and quality of hospital care. *N Engl J Med*. 2013;369(12):1134-1142.

21. Holmes DR Jr, Brennan JM, Rumsfeld JS, et al; STS/ACC TVT Registry. Clinical outcomes at 1 year following transcatheter aortic valve replacement. *JAMA*. 2015;313(10):1019-1028.

22. Murugiah K, Wang Y, Desai NR, Nuti SV, Krumholz HM. Hospital variation in outcomes for transcatheter aortic valve replacement among Medicare beneficiaries, 2011 to 2013. *J Am Coll Cardiol*. 2015;66(23):2678-2679.

23. Hannan EL, Samadashvili Z, Jordan D, et al. Thirty-day readmissions after transcatheter aortic valve implantation versus surgical aortic valve replacement in patients with severe aortic stenosis in New York state. *Circ Cardiovasc Interv*. 2015;8(8):e002744.

24. Panchal HB, Ladia V, Amin P, et al. A meta-analysis of mortality and major adverse cardiovascular and cerebrovascular events in patients undergoing transfemoral versus transapical transcatheter aortic valve implantation using Edwards valve for severe aortic stenosis. *Am J Cardiol*. 2014;114(12):1882-1890.

25. Biancari F, Rosato S, D'Errigo P, et al; OBSERVANT Research Group. Immediate and intermediate outcome after transapical versus

© 2017 American Medical Association. All rights reserved.
Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026

transfemoral transcatheter aortic valve replacement. *Am J Cardiol*. 2016;117(2):245-251.

26. van der Boon RM, Marcheix B, Tchetche D, et al. Transapical versus transfemoral aortic valve implantation: a multicenter collaborative study. *Ann Thorac Surg*. 2014;97(1):22-28.

27. Li X, Kong M, Jiang D, Dong A. Comparison 30-day clinical complications between transfemoral versus transapical aortic valve replacement for aortic stenosis: a meta-analysis review. *J Cardiothorac Surg*. 2013;8:168.

28. Siontis GC, Jüni P, Pilgrim T, et al. Predictors of permanent pacemaker implantation in patients with severe aortic stenosis undergoing TAVR: a meta-analysis. *J Am Coll Cardiol*. 2014;64(2):129-140.

29. Joynt KE, Jha AK. Thirty-day readmissions: truth and consequences. *N Engl J Med*. 2012;366(15):1366-1369.

30. Stub D, Lauck S, Lee M, et al. Regional systems of care to optimize outcomes in patients undergoing transcatheter aortic valve replacement. *JACC Cardiovasc Interv*. 2015;8(15):1944-1951.

31. Minha S, Waksman R, Satler LP, et al. Learning curves for transfemoral transcatheter aortic valve replacement in the PARTNER-I trial: success and safety. *Catheter Cardiovasc Interv*. 2016;87(1):165-175.

32. Alli OO, Booker JD, Lennon RJ, Greason KL, Rihal CS, Holmes DR Jr. Transcatheter aortic valve implantation: assessing the learning curve. *JACC Cardiovasc Interv*. 2012;5(1):72-79.

33. van der Boon RM, Nuis RJ, Benitez LM, et al. Frequency, determinants and prognostic implications of infectious complications after transcatheter aortic valve implantation. *Am J Cardiol*. 2013;112(1):104-110.

34. Tirado-Conte G, Freitas-Ferraz AB, Nombela-Franco L, et al. Incidence, causes, and impact of in-hospital infections after transcatheter aortic valve implantation. *Am J Cardiol*. 2016;118(3):403-409.

Editor's Note

The Balance Between Access and Quality in Transcatheter Valve Therapies

Patrick T. O'Gara, MD

Transcatheter aortic valve replacement (TAVR) is a transformative technology that has revolutionized the care of patients with symptomatic severe aortic stenosis. Procedural TAVR volumes are now comparable with those for isolated surgical AVR and continued growth is expected with expanding numbers of patients meeting eligibility requirements. Randomized clinical trials of TAVR vs surgical AVR in low surgical risk patients are under way; planning for a trial of TAVR vs active surveillance in asymptomatic patients is in its late stages. Currently, there are more than 500 US TAVR hospitals, a more than 3-fold increase since 2012.

Editorial page 721

Related article page 732

The rational dispersion of this technology has been a consistent concern since it was first approved for commercial use. How should patient access to this life-changing procedure be balanced against the need to promote the highest-quality care and insure optimal outcomes? Strategies used to address this challenge have included linkage of site selection criteria to Centers for Medicare & Medicaid Services reimbursement, US Food and Drug Administration indications for use labeling, institutional and operator requirements recommended by professional societies, and medical device company support. Participation in the Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy Registry (NCT01737528) is a requisite. The cumulative experience with TAVR is now robust enough to carefully examine not only survival and functional outcomes, but also the processes of care and measures of efficiency and quality that define best practices. It is inevitable that the conversation will turn to questions such as whether there are too many, too few, or just the right number of individual hospitals and whether alternative models of integrated care across hospitals should be considered, especially as the field of transcatheter therapies continues to evolve.

There is an inverse association between TAVR hospital volume and in-hospital death, vascular complications, and bleeding.[1] In this issue of *JAMA Cardiology*, Khera and colleagues[2] describe an inverse association between TAVR hospital volumes and 30-day readmission rates. Despite the controversies surrounding the readmission rate metric, scrutiny of processes designed to optimize transitions of care should be promoted.

Reports of this nature can inform policy discussions regarding dissemination of high-impact expensive technologies. Prior efforts to limit high-risk surgical procedures to higher-volume hospitals/operators were met with vigorous pushback.[3] Nevertheless, there is widespread acknowledgment that a prospective, multistakeholder conversation regarding a health systems approach to treating patients with valve disease, much like that espoused by the Brain Attack Coalition for acute stroke,[4] is necessary.

Author Affiliations: Cardiovascular Division, Brigham and Women's Hospital, Boston, Massachusetts; Associate Editor, *JAMA Cardiology*.

Corresponding Author: Patrick T. O'Gara, MD, Cardiovascular Division, Brigham and Women's Hospital, 75 Francis St, Boston, MA 2115 (pogara@bwh.harvard.edu).

Conflict of Interest Disclosures: The author has completed and submitted the ICMJE Form for Disclosure of Potential Conflicts of Interest and none were reported.

1. Carroll JD, Vemulapalli S, Dai D, et al. The association between procedural experience for transcatheter aortic valve replacement and outcomes: insights from the STS/ACC TVT Registry. *J Am Coll Cardiol*. In press.

2. Khera S, Kolte D, Gupta T, et al. Association between hospital volume and 30-day readmissions following transcatheter aortic valve replacement [published online May 11, 2017]. *JAMA Cardiol*. doi:10.1001/jamacardio.2017.1630

3. Urbach DR. Pledging to eliminate low-volume surgery. *N Engl J Med*. 2015;373(15):1388-1390.

4. Alberts MJ, Wechsler LR, Jensen ME, et al. Formation and function of acute stroke-ready hospitals within a stroke system of care recommendations from the brain attack coalition. *Stroke*. 2013;44(12):3382-3393.

© 2017 American Medical Association. All rights reserved.

Downloaded from jamanetwork.com by Cedars-Sinai Medical Library user on 03/27/2026